

Biological interfaces

A biological interface is a technology that connects a biological system (such as a human's muscle) to a digital system (such as a computer). The interface is the point where the two meet.

How computers help

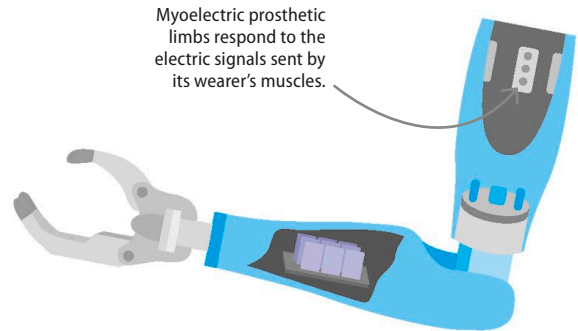
Digital systems are simple and straightforward. By contrast, biological systems have evolved over thousands of years by trial and error, and while they're very effective, they are not straightforward. When something in a human body breaks down, such as a weak heart or bad eyes, it can be extremely difficult to fix. Digital technologies aren't a perfect substitute, but they're relatively easy to produce and control. We can use them to tweak, correct, and enhance human organs. The biggest challenge is understanding the biological system well enough to interact with it.

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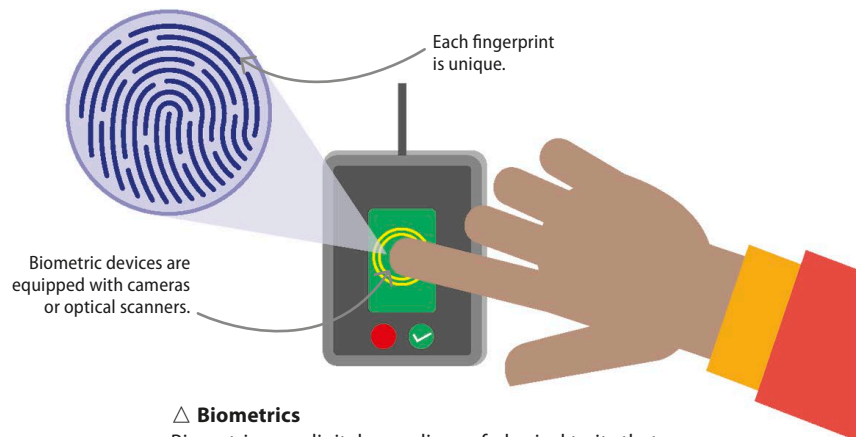
◀ 220–221 Computer science and disabilities

◀ 224–225 Predicting the future



△ Prosthetics

Prosthetics are used to replace lost arms, legs, and teeth, but can also be used to replace hearts. They range from motor-powered machines to metal rods with hinges. 3D printing technology has made it possible to build sophisticated, flexible, and affordable high-quality prosthetics. This is especially significant in developing countries.



△ Biometrics

Biometrics are digital recordings of physical traits that can be used to identify a person. Examples include fingerprints, retina scans, voice recognition, and even a person's gait. Biometrics can be added to smartphones, ATMs, and locks. It's a more reliable system than keycards or passwords, but it raises the possibility of identity theft.

REAL WORLD

Biometric passports

Biometric passports are passports that come embedded with a chip that stores the biological information of the holder. Sometimes this is simply the passport holder's name, details, signature, and picture. In other countries, the information includes fingerprints or eye maps. Some airports, such as those in Zurich, Switzerland, and Hong Kong, China, are testing new facial recognition software for check-ins and boarding gates, hoping to make the process safer and more efficient.

